

# Scalable and Provable Biclique-Preserving Clustering: The Power of Counting-based Approaches

Longlong Lin  
Southwest University  
Chongqing, China  
longlonglin@swu.edu.cn

Zeli Wang\*  
Chongqing University of Posts and  
Telecommunications  
Chongqing, China  
zlwang@cqupt.edu.cn

Rong-Hua Li  
Beijing Institute of Technology  
Beijing, China  
lironghuabit@126.com

Xiaohai Dai  
HuaZhong University of Science and  
Technology  
Wuhan, China  
xhdai@hust.edu.cn

Li Ni  
Anhui University  
Hefei, China  
nili@ahu.edu.cn

Jin Zhao  
HuaZhong University of Science and  
Technology  
Wuhan, China  
zjin@hust.edu.cn

## Abstract

Bipartite graphs are widely used to model relationships between entities of different types, where vertices are divided into two disjoint sets. Biclique-preserving clustering is a fundamental operation that retrieves clusters with dense bicliques, enabling various emerging applications. However, existing methods either fail to accurately capture the unique properties of bipartite graphs or significantly overlook the informative higher-order biclique substructure, leading to compromised clustering quality. Additionally, existing methods are overly dependent on biclique enumeration, resulting in poor scalability. To address these challenges, we propose *ECRC*, a *simple* yet *provable* Edge-Centric Reweighting Clustering framework that provides strict approximation guarantees for any biclique. A key advantage of *ECRC* is its ability to leverage powerful counting instead of exhaustive enumeration, significantly reducing time and space complexity. To further improve efficiency, we propose several effective graph reduction strategies to eliminate the unqualified vertices and edges before calculating the edge-centric weight. Extensive experiments on five datasets show that our algorithms are more efficient and effective compared to six baselines.

## CCS Concepts

• **Mathematics of computing** → **Graph algorithms.**

## Keywords

Graph Clustering; Bipartite Graph; Higher-order Graph Clustering

### ACM Reference Format:

Longlong Lin, Zeli Wang, Rong-Hua Li, Xiaohai Dai, Li Ni, and Jin Zhao. 2026. Scalable and Provable Biclique-Preserving Clustering: The Power of Counting-based Approaches. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3774904.3792116>

\*Corresponding author



This work is licensed under a Creative Commons Attribution 4.0 International License. *WWW '26, Dubai, United Arab Emirates*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2307-0/2026/04

<https://doi.org/10.1145/3774904.3792116>

## 1 Introduction

Higher-order graph clustering (*HGC*) is an emerging technique for analyzing the internal nature of complex networks, which aims at identifying the target cluster with dense motifs rather than edges [6, 41]. Unlike dyadic edges, each motif is a significant and frequently occurring subgraph pattern that indicates the unique interaction behavior among vertices and represents some particular functions [31, 57] (e.g., in e-commerce networks, each biclique may indicate fraudulent order-placing behavior). Namely, each motif should be regarded as an indivisible whole for clustering, and edges that do not participate in any motif instance do not contribute to the clustering function. Therefore, *HGC* can improve the clustering quality by considering the more semantic motifs [6, 10, 49].

Despite the significant success of *HGC*, most existing approaches are mainly tailored to unipartite graphs where the entities belong to a single type. However, in many real-life applications, entities can be modeled as two different types, and relationships exist only between entities of different types. For example, in an actor-movie network (e.g., IMDB), where each vertex represents an actor or a movie, and an edge indicates the actor has a role in the movie. Such networks are typically named bipartite graphs [11, 24, 42]. Figure 1 illustrates a synthetic toy bipartite graph. Numerous motifs are proposed in bipartite graphs, such as butterfly [36, 43, 45, 47] and bi-triangle [33, 55]. Perhaps, the most fundamental and representative motif is  $(p, q)$ -biclique [52, 53, 58] due to the other motifs can be considered as a relaxed  $(p, q)$ -biclique for different  $p$  and  $q$ . In this paper, we study a novel problem, named biclique-preserving clustering (*BPC*), which aims to detect a cluster (i.e., subgraph)  $(X, Y)$  such that  $(X, Y)$  contains many  $(p, q)$ -bicliques and there are few  $(p, q)$ -bicliques cut  $(X, Y)$  (see Section 2 for details). For real-world applications such as product recommendation, *BPC* would like to identify a subgraph consisting of similar taste customers and their favorite products [19, 48]. Besides, on document-word bipartite graphs, by utilizing *BPC* to group documents and words, the underlying themes or categories within the documents can be effectively identified [7, 54]. Thus, detecting *BPC* enables us to reveal interesting higher-order clusters hidden in bipartite graphs.

One simple treatment for *BPC* is to regard a bipartite graph as a unipartite graph and adopt the seminal motif-based clustering framework, e.g., [6, 41, 60, 62, 63], to construct a weighted graph where the weight of each edge  $e$  is the number of  $(p, q)$ -bicliques  $e$  participates in, then execute the traditional spectral clustering/graph diffusion algorithm on the weighted graph. However, the results obtained by the framework are not theoretically guaranteed (Table 1). Recently, Lin et al. [26] proposed a peeling-based method to integrate the computation and partition of the weighted graph in an iterative algorithm, which can obtain the results with approximation guarantees. However, these naive solutions ignore the unique topological properties of bipartite graphs with two disjoint vertex sets, resulting in sub-par theoretical (or empirical) clustering qualities (Table 1 and Section 4). More importantly, they have prohibitively high time & space overheads to enumerate and materialize all  $(p, q)$ -bicliques (especially for the larger  $p$  and  $q$ ) due to combinatorial explosions. For instance, on a medium-sized graph Flickr (Table 2) with  $1.8 \times 10^5$  edges, the number of  $(2, 4)$ -bicliques can exceed  $2 \times 10^{10}$ . Thus, they are hard to scale to massive bipartite graphs with millions of edges. On the other hand, as mentioned in Section A.1, numerous bipartite graph clustering algorithms have been proposed in the literature, such as projection-based methods [39, 67] and co-clustering methods [5, 7]. Unfortunately, these methods are lower-order techniques that ignore the more informative higher-order substructures, resulting in unsatisfactory empirical clustering qualities (Section 4). Summing up, the state-of-the-art (SOTA) methods either dilute clustering qualities due to inadequate exploitation of unique bipartite graph topology and the informative higher-order substructures, or incur poor scalability due to expensive biclique enumeration. Consequently, devising provable and scalable algorithms for *BPC* remains a significant challenge.

To overcome this challenge, we devise a provable *BPC* clustering framework *ECRC*. Specifically, we observe that the primary obstacle to theoretical guarantees in the existing motif-based clustering framework is the unclear relationship between biclique cut (i.e.,  $cut_{pq}(H)$  in Equation 1) and edge-weighted cut (i.e.,  $\sum_{u \in H, v \notin H} W_{uv}$  in Section 3.1). One fundamental reason for the above-mentioned phenomenon is that during the construction of the weighted graph, the weights of edges are obtained by constructing hyperedges (i.e., bicliques), resulting in the loss of the original edge information (more details in Section 3.2.1). Based on these in-depth observations, we propose the novel *edge-centric* reweighting clustering framework, termed *ECRC*, which can preserve the original edge information. At a high level, *ECRC* specifies that an edge's weight is non-zero iff it exists as an edge in the original bipartite graph and participates in at least one motif instance (Definition 6). We emphasize that our *ECRC* has three remarkably striking advantages: (1) We provide a rigorous proof establishing both theoretically and empirically high-quality approximation guarantees for *ECRC* across *all* values of  $p$  and  $q$  (Theorem 3), presenting a novel theoretical contribution that is currently lacking in the existing literature. (2) Our edge-centric weighted graph is smaller, at  $O(n + m)$ , while the existing method is  $O(\min\{n^2, \binom{p+q}{2} |\mathbb{B}_{pq}|\} + n)$ , in which  $n$  and  $m$  are the number of vertices and edges of the bipartite graph,  $|\mathbb{B}_{pq}|$  is the number of  $(p, q)$ -bicliques. Therefore, *ECRC* needs less time and space to partition the weighted graph. (3) Unlike existing methods

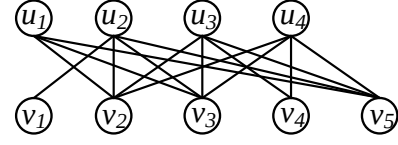


Figure 1: A synthetic toy bipartite graph.

that exhaustively enumerate all  $(p, q)$ -bicliques, *ECRC* can use the powerful counting-based techniques to directly obtain the edge weights in a combinatoric way, significantly reducing the computational cost. To further boost efficiency, we devise several effective graph reduction strategies to eliminate the unqualified vertices and edges before calculating the edge-centric weight. More concretely, we non-trivially use the  $(p, q)$ -core [9, 30] to delete the unqualified nodes and  $k$ -bitruss [37, 46] to remove the unpromising edges. Finally, we evaluate our solutions against six baselines on five datasets. Our solutions achieve a 1~4 orders of magnitude speedup with better clustering quality, while reducing memory usage by up to an order of magnitude.

## 2 Preliminaries

Let  $G = (L, R, E)$  be an unweighted and undirected bipartite graph, where  $L$  and  $R$  are two disjoint vertex sets (i.e.,  $L \cap R = \emptyset$ ), and  $E \subseteq L \times R$  is the edge set of  $G$ . We use  $n = |L \cup R|$  and  $m = |E|$  to denote the number of vertices and edges of  $G$ . For each vertex  $u \in L \cup R$ , we use  $N_G(u) = \{v | (u, v) \in E\}$  to denote the neighbors of  $u$  and  $d_G(u) = |N_G(u)|$  to represent the degree of  $u$ . Given a pair  $(L', R')$  with  $L' \subseteq L$  and  $R' \subseteq R$ , we use  $G(L', R') = (L', R', E')$  to denote the induced subgraph by  $(L', R')$ , where  $E' = \{(u, v) \in E | u \in L', v \in R'\}$ . If the context is clear, we use  $N(u)$ ,  $d(u)$ , and  $(L', R')$  to represent  $N_G(u)$ ,  $d_G(u)$ , and  $G(L', R')$ , respectively.

**DEFINITION 1.** ( $(p, q)$ -biclique [52, 53, 58]) Given an unweighted and undirected bipartite graph  $G$ , and two positive integer parameters  $p$  and  $q$ , a  $(p, q)$ -biclique is a complete subgraph  $B = G(P, Q)$ , satisfying (1)  $|P| = p$  and  $|Q| = q$ ; (2)  $\forall u \in P$  and  $\forall v \in Q$ ,  $(u, v) \in E$ . We use  $\mathbb{B}_{pq}$  to denote the set of all  $(p, q)$ -bicliques in  $G$ .

**EXAMPLE 1.** Figure 1 is a synthetic bipartite graph with 9  $(2, 3)$ -bicliques. For example,  $(\{u_3, u_4\}, \{v_2, v_3, v_4\})$ ,  $(\{u_3, u_4\}, \{v_2, v_4, v_5\})$ , and  $(\{u_3, u_4\}, \{v_3, v_4, v_5\})$  are  $(2, 3)$ -biclique. The rest of  $(2, 3)$ -bicliques are  $(\{u_1, u_2, u_3, u_4\}_2, \{v_2, v_3, v_5\})$ , in which  $\{u_1, u_2, u_3, u_4\}_2$  indicates choose arbitrary 2 nodes from  $\{u_1, u_2, u_3, u_4\}$ .

**DEFINITION 2.** (biclique-preserving graph clustering [6, 41, 62]) For an unweighted and undirected bipartite graph  $G$ , and two positive integer parameters  $p$  and  $q$ , the problem of the biclique-preserving graph clustering aims to find a high-quality cluster (i.e., induced subgraph)  $G(X, Y)$ , satisfying (1)  $G(X, Y)$  contains many  $(p, q)$ -bicliques; (2) there are few  $(p, q)$ -bicliques cross  $G(X, Y)$  and  $G \setminus G(X, Y)$ .

Based on the intuition of Definition 2, we propose the following biclique conductance to measure the clustering quality of an identified cluster  $G(X, Y)$ , which generalizes the well-known motif conductance [6, 41] from unipartite to bipartite graphs.

**DEFINITION 3.** (biclique conductance) For an unweighted and undirected bipartite graph  $G = (L, R, E)$ , and two positive integer parameters  $p$  and  $q$ , the biclique conductance of  $G(X, Y)$  is defined

as  $\phi_{pq}(X, Y) = \frac{cut_{pq}(H)}{\min\{vol_{pq}(H), vol_{pq}(\bar{H})\}}$ , in which  $H = X \cup Y$  and  $\bar{H} = (L \cup R) \setminus H$ .

$$cut_{pq}(H) = |\{B \in \mathbb{B}_{pq} | B \cap H \neq \emptyset, B \cap \bar{H} \neq \emptyset\}| \quad (1)$$

$$vol_{pq}(H) = \sum_{u \in H} |\{B \in \mathbb{B}_{pq} | u \in B\}| \quad (2)$$

By Definition 3, we can see that  $cut_{pq}(H)$  represents the number of  $(p, q)$ -bicliques that include at least one vertex from  $H$  and at least one vertex from  $\bar{H}$ . Meanwhile,  $vol_{pq}(H)$  (and  $vol_{pq}(\bar{H})$ ) indicates the number of  $(p, q)$ -bicliques that involve vertices from  $H$  (or  $\bar{H}$ , respectively). Consequently, edges that do not belong to any  $(p, q)$ -biclique do not contribute to the biclique conductance. This means that a cluster with numerous edges but few  $(p, q)$ -bicliques might exhibit poor biclique conductance. Thus, biclique conductance provides strong interpretability and enhances the clustering quality by emphasizing the  $(p, q)$ -bicliques, which are crucial structures.

**Remark.** When  $p = q = 1$ , a  $(p, q)$ -biclique corresponds to an edge, and the biclique conductance reduces to the classic edge-based conductance [16, 28]. When  $p = q = 2$ , the biclique conductance aligns with the butterfly-based higher-order graph clustering approach proposed by Zheng et al. [61]. Thus, our model serves as a generalization of the works referenced in [16, 28, 61].

**Problem Statement.** Given an unweighted and undirected bipartite graph  $G = (L, R, E)$ , and two positive integer parameters  $p$  and  $q$ . The goal of biclique-preserving graph clustering is to find a cluster  $(X^*, Y^*)$ , satisfying  $\phi_{pq}(X^*, Y^*) \leq \phi_{pq}(X, Y)$  for any  $X \subseteq L$  and  $Y \subseteq R$ . For convenience, we use  $\phi_{pq}^*$  to stand for  $\phi_{pq}(X^*, Y^*)$ .

Our problem is classified as NP-hard due to the NP-hardness of the traditional edge-based conductance [16, 18, 25, 27–29]. Consequently, following [6, 26, 41, 60, 62, 63], we also aim to devise approximation algorithms to mitigate the complexity without sacrificing practical clustering quality.

### 3 Methodologies

This section first proposes several basic solutions grounded in prevalent high-order graph clustering techniques. Then, we introduce a novel edge-centric clustering framework, distinguished by its strong approximation performance and exceptional scalability. Finally, several effective graph reduction strategies are proposed for further boost the efficiency.

#### 3.1 Basic Solutions and Their Limitations

To our knowledge, biclique-preserving clustering is a novel problem that has not been fully explored. A naive approach is to treat the bipartite graph as a unipartite graph and employ the existing high-order graph clustering techniques [6, 26, 41, 60, 62, 63] to partition the unipartite graph. To this end, we propose several basic solutions, which are briefly stated as follows (details in Section A).

**3.1.1 Order-based Solutions.** Let  $\mathcal{G}^{\mathbb{B}_{pq}} = (L \cup R, \mathcal{E}^{\mathbb{B}_{pq}})$  be the biclique-based unipartite weighted graph,  $\mathcal{E}^{\mathbb{B}_{pq}} = \{(u, v, W_{uv}) \mid u, v \in L \cup R\}$  represents the weighted edge set and  $W_{uv}$  denotes the number of  $(p, q)$ -bicliques containing both  $u$  and  $v$ . It is important to note that  $u$  and  $v$  may not necessarily be connected by an edge in the original graph (i.e.,  $(u, v) \notin E$ ). Based on these symbols, we can summarize the order-based solutions as follows: (1)

Sort all vertices according to a pre-defined rule. Without loss of generality, we denote the sorted vertices as  $u_1, u_2, \dots, u_n$ . (2) Output  $S = \arg \min \phi_{\mathcal{G}^{\mathbb{B}_{pq}}}(S_i)$ , where  $S_i = \{u_1, u_2, \dots, u_i\}$  and  $\phi_{\mathcal{G}^{\mathbb{B}_{pq}}}(S_i)$  represents the traditional edge-based conductance [28, 40] of  $S_i$  on  $\mathcal{G}^{\mathbb{B}_{pq}}$ . Thus, the well-known two order-based solutions are proposed: *BFV* [6] and *BPP* [60]. In particular, *BFV* and *BPP* employ the Fiedler Vector and Personalized PageRank vector, respectively, to sort all vertices. For convenience, Table 1 summarizes these basic solutions for comparison.

**Limitations of order-based solutions.** By Table 1, we have the following observations: (1) Although *BFV* can obtain the cluster with quadratic bound *only* when  $p + q = 3$ , the case of  $p + q = 3$  has no practical application for the biclique, as it generally requires  $p \geq 2$  and  $q \geq 2$  [52, 53, 58]. (2) *BPP* is heuristic algorithms, and their clustering qualities are significantly influenced by many hard-to-tune parameters and seeding strategies. In practical applications, its performance can be unstable and may even yield very poor solutions, as evidenced by our experimental results. (3) *BFV* and *BPP* entail excessively high costs, primarily because they require enumerating and storing all  $(p, q)$ -bicliques.

**3.1.2 The Peeling-based Solution.** Recently, Lin et al. [26] introduced a peeling-based approach that integrates the computation and partitioning of  $\mathcal{G}^{\mathbb{B}_{pq}}$  in an iterative manner, providing results with approximation guarantees. In particular, they redefine motif conductance, of which biclique conductance is a specific case with the motif is a biclique, and present a peeling-based greedy algorithm that iteratively optimizes motif conductance starting from each vertex. To fit our problem, we give the following definitions.

**DEFINITION 4. (biclique degree)** Consider a bipartite graph  $G$  and two positive integer parameters  $p$  and  $q$ . The biclique degree of a vertex  $u$  is defined as  $\mathbb{B}(u) = |\{B \in \mathbb{B}_{pq} \mid u \in B\}|$ , representing the number of bicliques in  $\mathbb{B}_{pq}$  that contain  $u$ . For a positive integer  $1 \leq j \leq p + q$  and a vertex subset  $C$ , we define  $\mathbb{B}_j^C(u) = |\{B \in \mathbb{B}_{pq} \mid u \in B, |C \cap B| = j\}|$ , which counts the number of bicliques in  $\mathbb{B}_{pq}$  that include  $u$  and intersect with  $C$  in exactly  $j$  vertices.

**DEFINITION 5. (biclique resident)** Consider a bipartite graph  $G$ , two positive integer parameters  $p$  and  $q$ , and a vertex subset  $C$ , the biclique resident of a vertex  $u \in C$  with respect to  $C$  is defined as  $Br_C(u) = \frac{\mathbb{B}(u) + \mathbb{B}_{p+q}^C(u) - \mathbb{B}_1^C(u)}{\mathbb{B}(u)}$ .

Based on these definitions, the biclique-preserving peeling removal (*BPR*) algorithm is outlined as follows: (1) Initially, it computes the biclique resident for every vertex. (2) Then, it iteratively removes the vertex with the smallest biclique resident, a step termed as *peeling*. (3) Finally, it produces the result with the smallest biclique conductance during the peeling process.

**Limitations of the peeling-based solution.** By Table 1, we know that while *BPR* provides results with approximation guarantees, it compromises the distinctive topological characteristics of bipartite graphs with two disjoint vertex sets, leading to suboptimal clustering quality, as stated in our empirical evaluation (Section 4). Besides, *BPR* incurs prohibitively high time costs due to the need to enumerate all  $(p, q)$ -bicliques, especially for larger values of  $p$  and  $q$ , resulting in a combinatorial explosion.

**Table 1: A comparison of biclique-preserving graph clustering methods.**  $p + q = 2$  is not discussed, because in this situation, biclique conductance (Definition 3) degenerates into the traditional edge-based conductance, which is a lower-order graph clustering and not the focus of this paper.  $\times$  represents the corresponding method has no accuracy guarantee.  $\phi_{pq}^* \in (0, 1]$  is the smallest biclique conductance.  $O(\text{Enu}(B_{pq}))$  (resp.,  $O(\text{Cou}(B_{pq}))$ ) is the time complexity of enumeration (resp., counting)  $(p, q)$ -biclique instances.  $|\mathbb{B}_{pq}|$  is the number of  $(p, q)$ -biclique.  $O(\text{Enu}(B_{pq}))$  is much slower than  $O(\text{Cou}(B_{pq}))$  [34, 52, 53, 58, 59].

| Methods    | Accuracy Guarantee                                                 | Time Complexity         | Space Complexity                                      | Remark                        |
|------------|--------------------------------------------------------------------|-------------------------|-------------------------------------------------------|-------------------------------|
| BFV [6]    | $O(\sqrt{\phi_{pq}^*})$ for $p + q = 3$ ; $\times$ for $p + q > 3$ | $O(\text{Enu}(B_{pq}))$ | $O(\min\{n^2, \binom{p+q}{2} \mathbb{B}_{pq} \} + m)$ | Biclique-based weighted graph |
| BPP [60]   | $\times$                                                           | $O(\text{Enu}(B_{pq}))$ | $O(\min\{n^2, \binom{p+q}{2} \mathbb{B}_{pq} \} + m)$ | Biclique-based weighted graph |
| BPR [26]   | $O(1/2 + 1/2\phi_{pq}^*)$                                          | $O(\text{Enu}(B_{pq}))$ | $O(m)$                                                | Biclique-based weighted graph |
| ECRC (Our) | $O(\sqrt{\phi_{pq}^*})$                                            | $O(\text{Cou}(B_{pq}))$ | $O(m)$                                                | Edge-centric weighted graph   |

### 3.2 Our Edge-centric Clustering Framework

As discussed in Section 3.1, these basic algorithms fail to deliver provably approximate results and often yield subpar practical outcomes due to their neglect of the unique characteristics of bipartite graphs. Besides, their reliance on biclique enumeration leads to poor scalability. To address these limitations, we propose an advanced Edge-Centric Reweighting Clustering framework, *ECRC*, which can produce both *provably and empirically high-quality* clusters. A significant advantage of *ECRC* is its ability to be implemented using the biclique counting method, making it faster than existing enumeration-based algorithms (e.g., *BFV* [6], *BPP* [60], *BPR* [26]).

**3.2.1 Novel Edge-Centric Weighted Graph.** Our method also utilizes the reweighting clustering framework proposed in Science 2016 [6] but makes a *simple yet very important* modification, which comes from the following in-depth observation.

**An in-depth observation.** According to Table 1, we can know that the reweighting clustering *BFV* is theoretically well-founded when  $p + q = 3$ . However, for the more general case where  $p + q > 3$ , *BFV* does not provide results with provable approximation guarantees. The main reason is that the relationship between  $\text{cut}_{pq}(H)$  and  $\sum_{u \in H, v \notin H} W_{uv}$  is unclear (see [6] for details). For example, whether  $\text{cut}_{pq}(H)$  can be bounded by  $\sum_{u \in H, v \notin H} W_{uv}$ , which is an open problem [6]. The intuitive explanation behind this is that the existing reweighting clustering first constructs a hypergraph where the hyperedges are all  $(p, q)$ -bicliques. Then, it transforms the hypergraph into a weighted graph using the clique expansion technique (where the weight of an edge  $(u, v)$  corresponds to the number of the vertex pair  $\{u, v\}$  participates in the hyperedges), which results in the loss of the original edge information (more details in Section 3.1). Therefore, an important question is whether we can construct a new weighted graph such that the relationship between  $\text{cut}_{pq}(H)$  and  $\sum_{u \in H, v \notin H} W_{uv}$  can be analyzed, while the relationship between  $\text{vol}_{pq}(H)$  and  $\sum_{u \in H, v \in L \cup R} W_{uv}$  can also be examined. Here, we provide a positive answer to this question by introducing a new edge-centric reweighting clustering framework, termed *ECRC*, which can preserve the original edge information. Based on these in-depth observations, we present the following novel edge-centric weighted graph.

**DEFINITION 6. (edge-centric weighted graph)** Given a bipartite graph  $G = (L, R, E)$ , and two positive integer parameters  $p$  and  $q$ , the edge-centric weighted graph is denoted as  $G_{ec}^{pq} = (L \cup R, W_{ec}^{pq})$ , where

$$(W_{ec}^{pq})_{uv} = \begin{cases} |\{B \in \mathbb{B}_{pq} | (u, v) \in B\}|, & (u, v) \in E \\ 0, & (u, v) \notin E \end{cases} \quad (3)$$

By Definition 6, we can know that if  $(u, v) \notin E$ , their edge-centric weight is 0. In other words, an edge in the edge-centric weighted graph has a non-zero weight iff it exists as an edge in the original bipartite graph and participates in at least one motif instance. Meanwhile, the edge-centric weighted graph has the following three-folds advantages: (1) We can obtain the practical high-quality result with provable approximation guarantees for any biclique (Theorem 3), a novel theoretical contribution absent from existing literature. (2) In the worst case scenario, the size of the edge-centric weighted graph  $G_{ec}^{pq}$  is smaller, at  $O(n + m)$ , while the biclique-based weighted graph  $G^{pq}$  is  $O(\min\{n^2, \binom{p+q}{2}|\mathbb{B}_{pq}|\} + n)$ . As a result, we can use less time and space to partition  $G_{ec}^{pq}$ . (3) We can implement *ECRC* through powerful counting techniques without exhaustively enumerating all  $(p, q)$ -bicliques, significantly reducing the computational cost. Next, we utilize the unique property of the bipartite graph, which has two disjoint vertex sets, to non-trivially derive the following significant lemmas, which lay a solid theoretical groundwork for the subsequent algorithm design.

**LEMMA 1.** Given an unweighted and undirected bipartite graph  $G = (L, R, E)$ , two positive integer parameters  $p$  and  $q$ , and a cluster  $G(X, Y)$  with  $H = X \cup Y$ , we have

$$\min\{p, q\} \text{cut}_{pq}(H) \leq \sum_{u \in H, v \notin H} (W_{ec}^{pq})_{uv} \leq pq \text{cut}_{pq}(H) \quad (4)$$

**LEMMA 2.** Given an unweighted and undirected bipartite graph  $G = (L, R, E)$ , two positive integer parameters  $p$  and  $q$ , and a cluster  $G(X, Y)$  with  $H = X \cup Y$ , we have

$$\min\{p, q\} \text{vol}_{pq}(H) \leq \sum_{u \in H} \sum_{v \in L \cup R} (W_{ec}^{pq})_{uv} \leq \max\{p, q\} \text{vol}_{pq}(H) \quad (5)$$

**Remark.** By Lemma 1 and Lemma 2, it follows that the proposed edge-centric weighted graph  $W_{ec}^{pq}$  can approximate both the  $\text{vol}_{pq}(\cdot)$  and  $\text{cut}_{pq}(\cdot)$  of the original graph, which constitutes our main technical contribution. Consequently, applying spectral clustering to  $W_{ec}^{pq}$  provides provable approximation guarantees (details in Theorem 3). However, existing methods ignore the unique properties of bipartite graphs (i.e., the bipartite graph has two disjoint vertex sets), leading to inaccurate estimates and subpar clustering quality.

**3.2.2 Provable Biclique-preserving Clustering.** Based on the above analyses, we devise a novel edge-centric reweighting clustering framework, which is outlined in Algorithm 1. Specifically, for any edge  $(u, v) \in G$ , Algorithm 1 first obtains the number of  $(p, q)$ -bicliques that contain  $(u, v)$  (Lines 2-3). Then, Algorithm 1 obtains the normalized Laplacian matrix  $\mathcal{L}$  by the previous  $(p, q)$ -biclique

**Algorithm 1** Edge-Centric Reweighting Clustering (ECRC)

---

**Input:** A bipartite  $G$  and two parameters  $p$  and  $q$   
**Output:** A biclique-preserving cluster  $S$

- 1: Initializing  $W_{ec}^{pq} \in R^{n \times n}$  and  $D_{ec}^{pq} \in R^{n \times n}$  are zero matrices
- 2: **for** each edge  $(u, v) \in G$  **do**
- 3:    $(W_{ec}^{pq})_{uv} \leftarrow$  the number of  $(p, q)$ -bicliques containing  $(u, v)$
- 4:    $(W_{ec}^{pq})_{vu} = (W_{ec}^{pq})_{uv}$
- 5:    $(D_{ec}^{pq})_{uu} += (W_{ec}^{pq})_{uv}; (D_{ec}^{pq})_{vv} += (W_{ec}^{pq})_{uv}$
- 6:  $\mathcal{L} \leftarrow I - (D_{ec}^{pq})^{-1/2} W_{ec}^{pq} (D_{ec}^{pq})^{-1/2}$
- 7:  $x \leftarrow$  eigenvector of the second smallest eigenvalue of  $\mathcal{L}$
- 8:  $x_i \leftarrow$  to be index of  $x$  with  $i$ th largest value
- 9:  $S \leftarrow \arg \min \phi_{W_{ec}^{pq}}(S_i)$ , where  $S_i = \{x_1, x_2, \dots, x_i\}$  and  $\phi_{W_{ec}^{pq}}(S_i)$  is the traditional edge-based conductance of  $S_i$  in terms of  $\phi_{W_{ec}^{pq}}$
- 10: **if**  $|S| > |\bar{S}|$  **then**
- 11:   **return**  $\bar{S}$
- 12: **else**
- 13:   **return**  $S$

---

counting (Line 6). Subsequently, Algorithm 1 identifies the eigenvector  $x$  corresponding to the second smallest eigenvalue of  $\mathcal{L}$  to perform the sweep procedure (Lines 7-9). This involves sorting all entries of  $x$  in descending order such that  $x_1 \geq x_2 \geq \dots \geq x_n$ , and then selecting the set  $S = \arg \min \phi_{W_{ec}^{pq}}(S_i)$ , where  $S_i = \{x_1, x_2, \dots, x_i\}$  and  $\phi_{W_{ec}^{pq}}(S_i)$  represents the traditional edge-based conductance [28, 40] of  $S_i$  on the weighted graph  $W_{ec}^{pq}$ . Finally, Algorithm 1 outputs the smaller set between  $S$  and its complement  $\bar{S}$  (Lines 11-13) due to  $\phi_{W_{ec}^{pq}}(S) = \phi_{W_{ec}^{pq}}(\bar{S})$ .

**THEOREM 1.** *Given a bipartite graph  $G$  and two parameters  $p$  and  $q$ , the time complexity and space complexity of Algorithm 1 are  $O(\text{Cou}(B_{pq}))$  and  $O(m)$ , respectively.  $O(\text{Cou}(B_{pq}))$  is the time complexity of counting  $(p, q)$ -biclique.*

**Remark.** By Table 1, it is evident that existing methods rely on biclique enumeration algorithms to obtain the hyperedges (i.e., biclique instances) and then construct the biclique-based weighted graph for clustering, necessitating at least  $O(\text{Enu}(B_{pq}))$  time. In contrast, our proposed ECRC algorithm is rooted in the biclique counting approach, allowing it to directly compute edge weights in  $O(\text{Cou}(B_{pq}))$  time for clustering. Note that  $O(\text{Enu}(B_{pq}))$  is much slower than  $O(\text{Cou}(B_{pq}))$  [34, 52, 53, 58, 59]. Besides, ECRC demands only linear space complexity for storing simple data structures, as established in Theorem 1. This is unlike other algorithms (except for BPR [26]), which must store all  $(p, q)$ -bicliques. Thus, the proposed ECRC algorithm outperforms SOTA methods in terms of efficiency, as also stated in our empirical results.

**THEOREM 2.** *Given an unweighted and undirected bipartite graph  $G = (L, R, E)$ , two positive integer parameters  $p$  and  $q$ , we let  $\lambda_2$  be the second smallest eigenvalue of  $\mathcal{L}(W_{ec}^{pq}) = I - (D_{ec}^{pq})^{-1/2} W_{ec}^{pq} (D_{ec}^{pq})^{-1/2}$ , where  $D_{ec}^{pq} = \text{diag}(W_{ec}^{pq} \mathbf{1})$  and  $\mathbf{1}$  is the all-one vector. We have*

$$\frac{\lambda_2 \min\{p, q\}}{2pq} \leq \phi_{pq}^* \leq \frac{\max\{p, q\}}{\min\{p, q\}} \sqrt{2\lambda_2} \quad (6)$$

**THEOREM 3.** *Given an unweighted and undirected bipartite graph  $G = (L, R, E)$ , two positive integer parameters  $p$  and  $q$ , we let  $(X, Y)$*

*be the cluster outputted by Algorithm 1. We have*

$$\phi_{pq}(X, Y) = O(\sqrt{\phi_{pq}^*}) \quad (7)$$

**Remark.** As shown in Table 1, our method ECRC achieves asymptotically comparable quality to BFV. However, unlike BFV, which fails to provide quality guarantees when  $p + q > 3$ , ECRC ensures consistent quality for any  $p$  and  $q$ . On the other hand, comparing the quality of ECRC with BPR is more complex, as it depends on  $\phi_{pq}^*$ . Nonetheless, as illustrated in our empirical results in Section 4, ECRC consistently demonstrated superior performance, likely due to BPR being tailored for unipartite graphs and consequently overlooking the distinctive characteristics of bipartite graphs. Thus, the ECRC produces both provably and empirically high-quality clusters compared to SOTA methods.

### 3.3 Bipartite Graph Reduction

In this subsection, we utilize the unique property of the bipartite graph to present several effective graph reduction strategies, which can eliminate unqualified vertices and edges, leading to a substantial graph reduction before calculating the edge-centric weight.

**DEFINITION 7 (( $p, q$ )-CORE [9, 30]).** *Given a bipartite graph  $G$  and two positive integers  $p$  and  $q$ , the  $(p, q)$ -core of  $G$  is a maximal subgraph  $(X, Y)$  such that every node in  $X$  has at least  $q$  neighbors in  $Y$ , and every node in  $Y$  has at least  $p$  neighbors in  $X$ .*

**LEMMA 3.** *Let  $(X, Y)$  is the  $(p, q)$ -core, we can safely delete any node that is not in  $(X, Y)$ .*

**PROOF.** According to the definition of the  $(p, q)$ -biclique  $(P, Q)$ , we can know that every node  $u \in P$  must have at least  $q$  neighbors in  $Q$ , and every node  $v \in Q$  must have at least  $p$  neighbors in  $P$ . Thus, any node that is not in  $(X, Y)$  can be safely pruned.  $\square$

Lemma 3 can be efficiently executed with  $O(m)$  time complexity [9]. Next, we propose the bitruss model to prune unqualified edges.

**DEFINITION 8 ( $k$ -BITRUSS [37, 46]).** *Given a bipartite graph  $G$  and a positive integer  $k$ , the  $k$ -bitruss of  $G$  is a maximal subgraph  $(X, Y)$  such that every edge in  $(X, Y)$  is contained in at least  $k$  butterflies (i.e.,  $(2, 2)$ -bicliques) of  $(X, Y)$ .*

**LEMMA 4.** *Let  $(X, Y)$  is the  $(p - 1)(q - 1)$ -bitruss, we can safely delete any edge not in  $(X, Y)$ .*

**PROOF.** According to the definition of the  $(p, q)$ -biclique  $(P, Q)$ , we can know that every edge in  $(P, Q)$  must be contained in at least  $(p - 1)(q - 1)$  butterflies of  $(P, Q)$ . Thus, any edge that is not in  $(X, Y)$  can be safely pruned.  $\square$

Computing  $k$ -bitruss is not profitable because the benefit of its pruning effect is outweighed by the cost of computing  $k$ -bitruss. The main performance bottleneck of  $k$ -bitruss is the peeling phase [44], which incurs expensive edge removal operations. To balance efficiency and pruning effectiveness, we can use near-linear time complexity to calculate the number (denoted as  $\mathbb{B}_{22}(uv)$ ) of butterflies each edge  $(u, v)$  participates in. Namely, if  $\mathbb{B}_{22}(uv) < (p - 1)(q - 1)$ , we can safely prune  $(u, v)$  (Lemma 4). To our knowledge, the  $(p, q)$ -core is commonly used to eliminate unqualified vertices. However, we are the first to adopt the  $k$ -bitruss for pruning.

## 4 Experimental Evaluation

This section conducts extensive experiments to evaluate the efficiency and effective of the proposed solutions. All experiments are conducted on a Linux server with one Intel(R) Xeon(R) Silver 4210 2.20GHz CPU and 1 TB RAM.

### 4.1 Experimental Setup

**Datasets.** We evaluate our solutions using five real-world bipartite graphs (Table 2) from various domains, which are commonly used benchmarks for bipartite graph clustering [50, 54]. Specifically, the IMDB dataset represents an actor-movie network, where vertices correspond to actors or movies, and edges indicate that an actor has a role in a movie. Both Cora and Citeseer are document networks containing scientific publications and the corresponding keywords. The Flickr dataset is a social network where vertices represent users or tags, and edges capture users' interest in specific tags. The Google is extracted from Google Maps, with edges representing reviews of books posted by users.

**Algorithms.** The following algorithms are implemented for comparison. (1) Lower-order bipartite graph clustering: SBC [22], CCMOD [1], and InfoCC [8]. SBC is rooted in spectral clustering, whereas CCMOD and InfoCC are based on modularity optimization and information theory, respectively. (2) Higher-order unipartite graph clustering (Section 3.1): *BFV* [6], *BPP* [60], and *BPR* [26]. (3) Higher-order bipartite graph clustering: *ECRC* (Algorithm 1) and *ECRC-R*. *ECRC-R* is *ECRC* with the graph reduction strategies proposed in Section 3.3. We employ the SOTA BCList++ [52, 53] for enumerating and EPivoter [58] for counting  $(p, q)$ -bicliques, both of which are more general and efficient. Our main contribution lies not in accelerating biclique counting itself, but in the ingenious transformation of an enumeration-based framework into a novel counting-based clustering framework. So, any more efficient counting algorithm can be seamlessly incorporated into our approach.

**Parameters.** Unless specified otherwise, we take the default parameters of these baselines in our experiments. Specifically, SBC is implemented by the machine learning library scikit-learn. CCMOD and InfoCC are implemented by the CoClust package [35]. *BFV*, *BPP*, and *BPR* are implemented by the corresponding papers. Note that since *BPP* takes a seed vertex as input, to be more reliable, we randomly select 10 vertices as seed vertices and report the average running time and quality. In practical scenarios, the values of  $p$  and  $q$  are typically not excessively large due to the motif  $((p, q)$ -biclique is also a motif) is a small subgraph [52, 53, 58]. Thus, we generate multiple  $(p, q)$  pairs by setting  $p + q < 7$  [34, 52, 53, 58, 59].

### 4.2 Clustering Efficiency Evaluation

As the objective functions of lower-order bipartite clustering methods, such as SBC [22], CCMOD [1], and InfoCC [8], differ from the biclique conductance analyzed in this paper, comparing their efficiency is neither relevant nor essential.

**Exp-1: Running time of different algorithms with varying parameters.** The running time of *BFV*, *BPP*, *BPR*, and *ECRC* across all datasets with varying  $p$  and  $q$  is detailed in Figure 2. Notably, we omit empirical results for *ECRC-R* at this stage, as the impact of graph reduction techniques will be discussed later. Figure 2 reveals

**Table 2: Dataset statistics.**  $\bar{d}_L$  and  $\bar{d}_R$  are the average degree of  $L$  and  $R$ , respectively.

| Dataset       | $ L $  | $ R $   | $ E $     | $\bar{d}_L$ | $\bar{d}_R$ |
|---------------|--------|---------|-----------|-------------|-------------|
| IMDB [50]     | 617    | 1,398   | 5,322     | 8.6         | 3.8         |
| Cora [15]     | 2,708  | 1,432   | 49,216    | 18.2        | 34.4        |
| Citeseer [15] | 3,312  | 3,703   | 105,165   | 31.8        | 28.4        |
| Flickr [21]   | 7,564  | 12,047  | 182,517   | 24.1        | 15.1        |
| Google [51]   | 64,527 | 868,937 | 1,487,747 | 23.1        | 1.7         |

the following key observations: (1) *ECRC* consistently outperforms all other methods on all datasets. Its efficiency is attributed to its novel edge-centric reweighting clustering framework, which leverages an efficient biclique counting method. In contrast, *BFV*, *BPP*, and *BPR* rely on exhaustive enumeration of all biclique instances, making them significantly slower. Notably, *ECRC* achieves speedups of  $1 \sim 4$  orders of magnitude over these enumeration-based methods in most cases. For instance, on the Google dataset with  $p = 2$  and  $q = 4$ , *ECRC* finds the results in just 4 seconds, whereas *BFV*, *BPP*, and *BPR* require 125,829, 125,959, and 125,819 seconds, respectively. The running time of *ECRC* typically peaks when  $p = 2$  and  $q = 4$ . This is likely due to the decreased biclique counting costs under these parameter settings. (2) *ECRC* exhibits relatively stable performance as  $p + q$  increases, whereas the baseline methods show a significant rise in runtime. This discrepancy arises because larger  $p + q$  values force baseline methods to enumerate more bicliques, while *ECRC* only requires a simple combinatorial count (Table 1). As a result, these results show our *ECRC* is highly efficient.

**Exp-2: The pruning performance of graph reduction with varying parameters.** Figure 3 illustrates the impact of two graph reduction optimizations (introduced in Section 3.3) on the *ECRC* algorithm. For clarity, we refer to *ECRC* with  $(p, q)$ -core (Lemma 3) and  $k$ -bitruss (Lemma 4) as *ECRC-core* and *ECRC-truss*, respectively. The results show a consistent decrease in runtime with each additional optimization. On average, *ECRC-core* achieves a twofold speedup over *ECRC*, while *ECRC-truss* further improves performance by approximately three times. The fully optimized version *ECRC-R* delivers the best results. These findings highlight the effectiveness of the proposed graph reduction strategies.

**Exp-3: Scalability testing.** To further evaluate the scalability of our solutions, we generate extensive synthetic graphs. Figure 4 presents results for  $p = 3$  and  $q = 2$ , with similar trends observed across other values of  $p$  and  $q$ . Specifically, we sample either edges or nodes from the Google dataset, which has the maximum number of edges among the tested datasets, varying the sample percentage from 50% to 100%. As shown in Figure 4, our proposed algorithm *ECRC* demonstrates strong scalability with respect to graph size. In contrast, the runtime of other baselines fluctuates significantly as the graph size increases. This instability arises because their time complexity is nonlinear in terms of  $n$  or  $m$ , or even reaches  $n^{(p+q)}$  (as shown in Table 1). These results highlight that our algorithms scale efficiently on massive graphs, whereas the baselines do not.

**Exp-4: Memory overhead comparison.** Figure 5 illustrates the memory overhead of the evaluated algorithms for  $p = 3$  and  $q = 2$ , showing similar trends across other values of  $p$  and  $q$ . As expected, our proposed *ECRC* consistently outperforms all baselines in memory efficiency across all datasets, achieving up to an

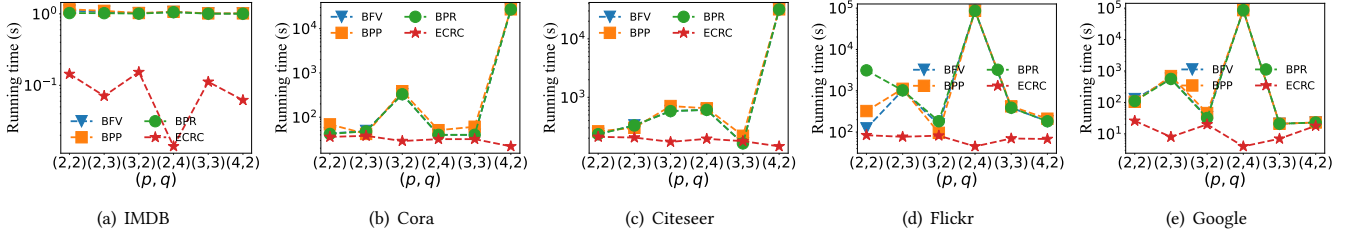


Figure 2: Running time of different algorithms with varying parameters.

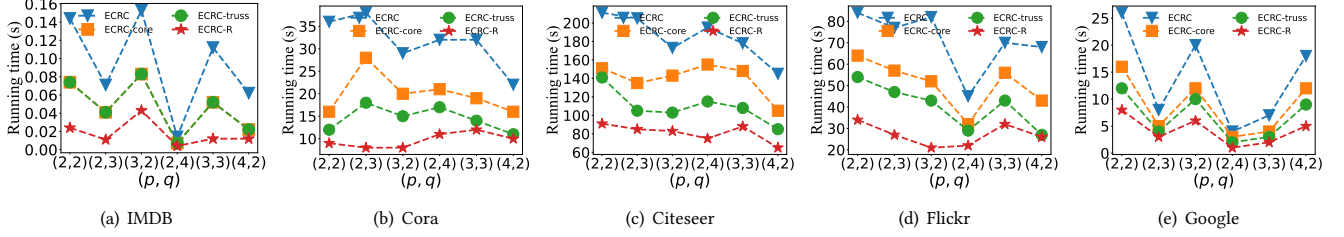


Figure 3: The pruning performance of graph reduction with varying parameters.

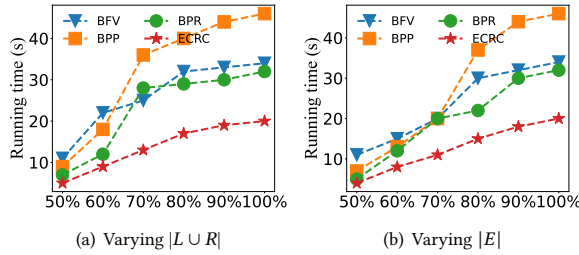


Figure 4: Scalability testing.

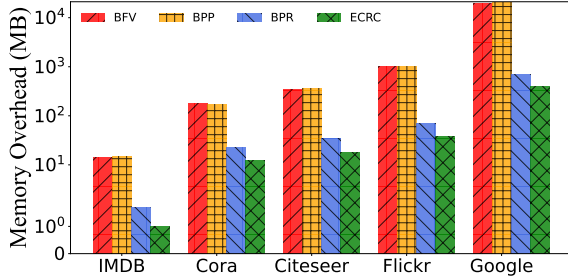


Figure 5: Memory overhead.

order-of-magnitude reduction compared to *BFV* and *BPP*. For example, on the Google dataset, *ECRC* consumes 389 MB, whereas *BFV* and *BPP* require 19,672 MB and 20,130 MB, respectively. This significant reduction stems from our novel edge-centric clustering framework, which only assigns weights to the original graph's edges, resulting in at most  $O(m)$  memory usage. In contrast, *BFV* and *BPP* assign edge weights by constructing hyperedges (where a hyperedge corresponds to a  $(p, q)$ -clique), leading to a memory complexity of  $O(\min\{n^2, \binom{p+q}{2} |B_{pq}|\} + m)$  (Table 1). Additionally, *BPR* incurs slightly higher memory overhead than *ECRC*. This is because *BPR* maintains some auxiliary structures, such as biclique resident information for each vertex (Algorithm 4). These results show the superior memory efficiency of our algorithm.

Table 3: Clustering quality of various clustering methods. The best and second-best results in each metric are marked in bold and underlined, respectively.

| BC/ACC      | IMDB                | Cora                | Citeseer            | Flickr                      | Google              |
|-------------|---------------------|---------------------|---------------------|-----------------------------|---------------------|
| SBC         | 0.132/0.081         | 0.342/0.294         | 0.274/0.204         | 0.208/0.326                 | 0.304/0.307         |
| CCMOD       | 0.215/0.123         | 0.238/0.361         | 0.313/0.157         | 0.328/0.187                 | 0.152/0.340         |
| InfoCC      | 0.174/0.094         | <u>0.126</u> /0.344 | 0.353/0.114         | 0.246/0.188                 | 0.292/0.342         |
| <i>BFV</i>  | <u>0.002</u> /0.653 | 0.393/0.298/        | 0.395/0.219         | <u>0.025</u> /0.491         | 0.003/0.329         |
| <i>BPP</i>  | <u>0.002</u> /0.624 | 0.420/0.421         | <u>0.445</u> /0.222 | 0.203/0.427                 | 0.008/0.323         |
| <i>BPR</i>  | <u>0.002</u> /0.629 | 0.198/0.604         | <u>0.232</u> /0.194 | 0.103/0.790                 | <u>0.002</u> /0.365 |
| <i>ECRC</i> | <b>0.000</b> /0.773 | <b>0.000</b> /0.667 | <b>0.194</b> /0.153 | <b>0.017</b> / <u>0.500</u> | <b>0.000</b> /0.545 |

### 4.3 Clustering Quality Evaluation

We evaluate the quality of the identified clusters using the proposed biclique conductance (BC), as defined in Definition 3. Besides, we also test quality using Clustering Accuracy (ACC) [2, 5, 65, 66], which quantifies the alignment between detected clusters and ground-truth clusters. A lower BC and higher ACC indicate better clustering quality. Note that we do not report traditional edge-based metrics (e.g., modularity or density) since they test cluster quality for lower-order clustering methods.

**Exp-5: Clustering quality of various clustering methods.** Since traditional lower-order bipartite graph clustering methods (i.e., SBC, CCMOD, and InfoCC) have no parameters  $p$  and  $q$ , Table 3 only shows results for  $p = 3$  and  $q = 2$ . The empirical results of other  $p$  and  $q$  will be discussed later. For the BC metric, the key findings are: (1) *ECRC* outperforms all other methods on all datasets. This superiority is attributed to *ECRC*'s ability to identify clusters with a near-linear approximation ratio, whereas other methods provide poor or no theoretical guarantees (Table 1). (2) *BPP*, *SBC*, *CCMOD*, and *InfoCC* exhibit poor clustering quality. This is because *BPP* is a heuristic method, lacks guarantees on clustering performance (Table 1), while *SBC*, *CCMOD*, and *InfoCC* are lower-order graph clustering methods that fail to account for essential higher-order biclique structures. For the ACC metric, the observations are: (1) *ECRC* achieves superior performance on three out of five datasets while achieving comparable performance on the remaining

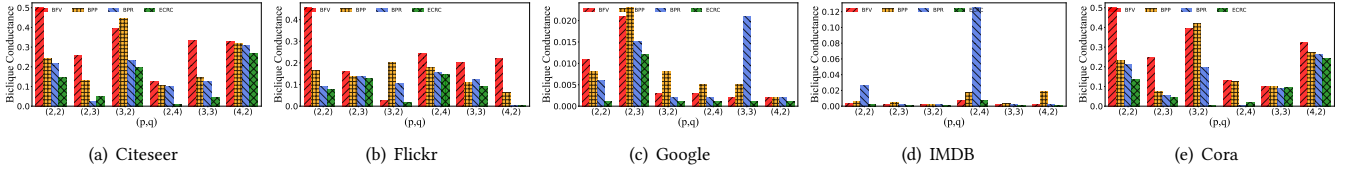


Figure 6: Effectiveness of different biclique-preserving clustering methods with varying parameters.

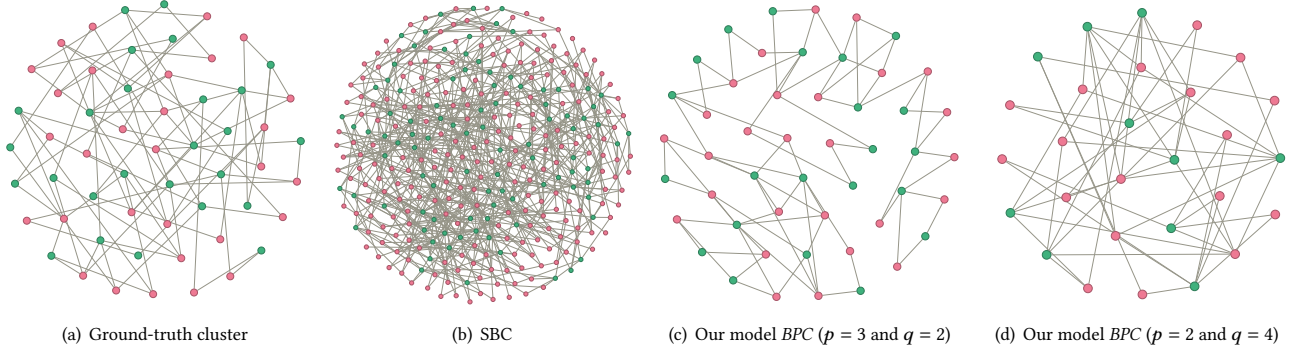


Figure 7: Case studies. Nodes of different colors represent different types of nodes.

two datasets. (2) *SBC*, *CCMOD*, and *InfoCC* demonstrate poor ACC across all datasets. This is because these traditional lower-order clustering methods fail to capture the higher-order biclique structural information essential for effective graph clustering. Overall, these findings show that our proposed solution consistently yields higher-quality clusters compared to baseline methods.

**Exp-6: Effectiveness of different biclique-preserving clustering methods with varying parameters.** By Figure 6, we have the following observations: (1) Our proposed *ECRC* outperforms other baselines across most cases under different values of  $p$  and  $q$ . (2) On average, *BFV* and *BPP* ranking as the worst and second worst, respectively. While *BPR*'s performance remains passable due to its theoretical guarantees (albeit not ideal), *BFV* and *BPP* rely on heuristics without strict theoretical guarantees. (3) The performance of these baselines varies significantly depending on the dataset. For instance, on the IMDB dataset, the performance order is  $BFV < BPP < BPR$  on average, whereas on the Cora dataset, the order is  $BPR < BPP < BFV$ , with  $A > B$  indicating that  $A$  has larger biclique conductance. These results highlight that our *ECRC* can identify higher-quality clusters than baseline methods on real-world graphs, aligning with our theoretical analysis (Table 1).

**Exp-7: Case Studies.** We conduct case studies on the IMDB dataset [50], where vertices represent actors or movies, and edges indicate that an actor has a role in a movie. To evaluate the utility of the proposed *BPC* (as *BFV*, *BPP*, and *BPR* share the same objective function as our *BPC*, we omitted them), we use the ground-truth cluster, which consists of 20 actors and 37 movies (as shown in Figure 7(a)). Figure 7 visualizes the clusters identified by different models. Notably, the clusters found by *CCMOD* and *InfoCC* are too large (close to 1000 nodes) to be shown in the figure, and thus are omitted. These models also exhibit poor clustering quality, as discussed in Table 3. In this instance, *SBC* identifies a cluster of 96 actors and 228 movies, which are densely connected by edges,

but not by bicliques (Figure 7(b)). A detailed analysis of Figure 7(b) reveals that *SBC* includes (or misses) numerous nodes that do not belong to the ground-truth cluster (or should be included), resulting in low clustering accuracy (see Table 3). In contrast, our model, *BPC*, with  $p = 3$  and  $q = 2$  (Figure 7(c)), correctly identifies nodes from the ground-truth cluster, with only a few ground-truth nodes undetected. This makes *BPC* highly compatible with the ground-truth cluster and achieves impressive clustering accuracy (see Table 3). Additionally, the cluster identified by *SBC* is densely connected by lower-order edges but lacks stable higher-order bicliques, leading to poor interpretability [6, 41, 60]. Other  $p$  and  $q$  settings also have similar trends, thus, we only visualize the result for  $p = 2$  and  $q = 4$  in Figure 7 (d) for brevity. In summary, these results underscore the potential of our model for effectively clustering bipartite graphs and related applications.

## 5 Conclusion

We first introduce a novel edge-centric clustering framework, *ECRC*, to identify biclique-preserving clusters in bipartite graphs. Notably, *ECRC* is *simple*, yet achieves *provable* approximation guarantees for any biclique. Moreover, *ECRC* can be implemented through combinatorial counting, making it significantly faster than existing enumeration-based methods. Subsequently, we propose several powerful graph reduction strategies to boost the efficiency of *ECRC*. Finally, we present extensive empirical results on five real-world datasets, comparing our approach against six competing methods to evaluate the efficiency and effectiveness of the proposed solutions.

## 6 ACKNOWLEDGMENTS

The work is supported in part by the National Natural Science Foundation of China under Grant 62402399, in part by the Natural Science Foundation of Chongqing, China under Grant CSTB2025NSCQ-GPX1268, and in part by the Science and Technology Research Program of Chongqing Municipal Education Commission under Grant KJQN202500647.

## References

- [1] Melissa Ailem, François Role, and Mohamed Nadif. 2015. Co-clustering Document-term Matrices by Direct Maximization of Graph Modularity. In *CIKM*. 1807–1810.
- [2] Sinan Aksoy, Tamara G. Kolda, and Ali Pinar. 2017. Measuring and modeling bipartite graphs with community structure. *J. Complex Networks* 5, 4 (2017), 581–603.
- [3] Noga Alon and Vitali D Milman. 1985.  $\lambda_1$ , isoperimetric inequalities for graphs, and superconcentrators. *Journal of Combinatorial Theory, Series B* 38, 1 (1985).
- [4] Alex Arenas, Alberto Fernandez, Santo Fortunato, and Sergio Gomez. 2008. Motif-based communities in complex networks. *Journal of Physics A: Mathematical and Theoretical* 41, 22 (2008), 224001.
- [5] Michael J Barber. 2007. Modularity and community detection in bipartite networks. *Physical Review E* 76, 6 (2007), 066102.
- [6] Austin R Benson, David F Gleich, and Jure Leskovec. 2016. Higher-order organization of complex networks. *Science* 353, 6295 (2016), 163–166.
- [7] Inderjit S. Dhillon. [n. d.]. Co-clustering documents and words using bipartite spectral graph partitioning. In *KDD*. 269–274.
- [8] Inderjit S. Dhillon, Subramanyam Mallela, and Dharmendra S. Modha. 2003. Information-theoretic co-clustering. In *kdd*, Lise Getoor, Ted E. Senator, Pedro M. Domingos, and Christos Faloutsos (Eds.). 89–98.
- [9] Danhao Ding, Hui Li, Zhipeng Huang, and Nikos Mamoulis. 2017. Efficient Fault-Tolerant Group Recommendation Using alpha-beta-core. In *CIKM*.
- [10] Alexandre Duval and Fragkiskos D. Malliaros. 2022. Higher-order Clustering and Pooling for Graph Neural Networks. In *CIKM*. 426–435.
- [11] David Eppstein. 1994. Arboricity and Bipartite Subgraph Listing Algorithms. *Inf. Process. Lett.* 51, 4 (1994), 207–211.
- [12] Yixiang Fang, Kaiqiang Yu, Reynold Cheng, Laks V. S. Lakshmanan, and Xuemin Lin. 2019. Efficient Algorithms for Densest Subgraph Discovery. *Proc. VLDB Endow.* 12, 11 (2019), 1719–1732.
- [13] Miroslav Fiedler. 1973. Algebraic connectivity of graphs. *Czechoslovak mathematical journal* 23, 2 (1973), 298–305.
- [14] Santo Fortunato and Marc Barthélemy. 2007. Resolution limit in community detection. *Proceedings of the national academy of sciences* 104, 1 (2007), 36–41.
- [15] Lise Getoor. 2005. *Link-based Classification*. Springer, 189–207.
- [16] David F. Gleich and C. Seshadhri. 2012. Vertex neighborhoods, low conductance cuts, and good seeds for local community methods. In *KDD*.
- [17] Gene H Golub and Charles F Van Loan. 2013. *Matrix computations*. JHU press.
- [18] Yue He, Longlong Lin, Pingpeng Yuan, Ronghua Li, Tao Jia, and Zeli Wang. 2024. CCSS: Towards conductance-based community search with size constraints. *Expert Syst. Appl.* 250 (2024), 123915.
- [19] Reinhard Heckel, Michail Vlachos, Thomas P. Parnell, and Celestine Dünner. 2017. Scalable and Interpretable Product Recommendations via Overlapping Co-Clustering. In *ICDE*. 1033–1044.
- [20] Ling Huang, Hong-Yang Chao, and Guangqiang Xie. 2020. MuMod: A Micro-Unit Connection Approach for Hybrid-Order Community Detection. In *AAAI*.
- [21] Xiao Huang, Jundong Li, and Xia Hu. 2017. Label Informed Attributed Network Embedding. In *WSDM*. 731–739.
- [22] Yuval Kluger, Ronen Basri, Joseph T Chang, and Mark Gerstein. 2003. Spectral biclustering of microarray data: coclustering genes and conditions. *Genome research* 13, 4 (2003), 703–716.
- [23] Lazhar Labiod and Mohamed Nadif. 2011. Co-clustering for Binary and Categorical Data with Maximum Modularity. In *ICDM*. 1140–1145.
- [24] Matthieu Latapy, Clémence Magnien, and Nathalie Del Vecchio. 2008. Basic notions for the analysis of large two-mode networks. *Soc. Networks* 30, 1 (2008), 31–48.
- [25] Longlong Lin, Yue He, Wei Chen, Pingpeng Yuan, Rong-Hua Li, and Tao Jia. 2025. Effective and Efficient Conductance-Based Community Search at Billion Scale. *IEEE Trans. Big Data* 11, 6 (2025), 3170–3184.
- [26] Longlong Lin, Tao Jia, Zeli Wang, Jin Zhao, and Rong-Hua Li. 2024. PSMC: Provable and Scalable Algorithms for Motif Conductance Based Graph Clustering. In *KDD*. 1793–1803.
- [27] Longlong Lin, Quanao Li, Miao Qiao, Zeli Wang, Jin Zhao, Rong-Hua Li, Xin Luo, and Tao Jia. 2026. NCSAC: Effective Neural Community Search via Attribute-Augmented Conductance. *IEEE Transactions on Knowledge and Data Engineering* 38, 2 (2026), 1221–1235.
- [28] Longlong Lin, Ronghua Li, and Tao Jia. 2023. Scalable and Effective Conductance-Based Graph Clustering. In *AAAI*. 4471–4478.
- [29] Longlong Lin, Zeli Wang, Rong-Hua Li, Qiyu Liu, Hongchao Qin, and Jin Zhao. 2025. Provable Higher-Order Graph Clustering: The Power of Peeling-Based Approaches. *IEEE Trans. Knowl. Data Eng.* 37, 9 (2025), 5568–5583.
- [30] Boge Liu, Long Yuan, Xuemin Lin, Lu Qin, Wenjie Zhang, and Jingren Zhou. 2019. Efficient  $(\alpha, \beta)$ -core Computation: an Index-based Approach. In *WWW*. ACM, 1130–1141.
- [31] R. Milo, S. Shen-Orr, S. Itzkovitz, N. Kashtan, D. Chklovskii, and U. Alon. 2002. Network motifs: simple building blocks of complex networks. *Science* (2002).
- [32] Mark E. Newman. 2004. Fast algorithm for detecting community structure in networks. *Physical Review E* (2004), 66–133.
- [33] Tore Opsahl. 2013. Triadic closure in two-mode networks: Redefining the global and local clustering coefficients. *Soc. Networks* 35, 2 (2013), 159–167.
- [34] Linshan Qiu, Zhonggen Li, Xiangyu Ke, Lu Chen, and Yunjun Gao. 2024. Accelerating biclique counting on GPU. In *ICDE*. 3191–3203.
- [35] François Role, Stanislas Morbieu, and Mohamed Nadif. 2019. Coclust: a python package for co-clustering. *Journal of Statistical Software* 88 (2019), 1–29.
- [36] Seyed-Vahid Sanei-Mehri, Ahmet Erdem Sariyüce, and Srikanta Tirathapura. 2018. Butterfly Counting in Bipartite Networks. In *KDD*. 2150–2159.
- [37] Ahmet Erdem Sariyüce and Ali Pinar. 2018. Peeling Bipartite Networks for Dense Subgraph Discovery. In *WSDM*. 504–512.
- [38] Binta Sun, Maximilian Danisch, T.-H. Hubert Chan, and Mauro Sozio. 2020. KClust++: A Simple Algorithm for Finding k-Clique Densest Subgraphs in Large Graphs. *Proc. VLDB Endow.* 13, 10 (2020), 1628–1640.
- [39] Raphael Tackx, Fabien Tarissan, and Jean-Loup Guillaume. 2017. ComSim: A Bipartite Community Detection Algorithm Using Cycle and Node’s Similarity. In *COMPLEX NETWORKS*, Vol. 689. 278–289.
- [40] Luca Trevisan. 2017. Lecture notes on graph partitioning, expanders and spectral methods. *University of California, Berkeley*, <https://people.eecs.berkeley.edu/luca/books/expanders-2016.pdf> (2017).
- [41] Charalampos E. Tsourakakis, Jakub Pachocki, and Michael Mitzenmacher. 2017. Scalable Motif-aware Graph Clustering. In *WWW*. 1451–1460.
- [42] Jun Wang, Arjen P. de Vries, and Marcel J. T. Reinders. 2006. Unifying user-based and item-based collaborative filtering approaches by similarity fusion. In *SIGIR*. 501–508.
- [43] Kai Wang, Xuemin Lin, Lu Qin, Wenjie Zhang, and Ying Zhang. 2019. Vertex Priority Based Butterfly Counting for Large-scale Bipartite Networks. *Proc. VLDB Endow.* 12, 10 (2019), 1139–1152.
- [44] Kai Wang, Xuemin Lin, Lu Qin, Wenjie Zhang, and Ying Zhang. 2022. Towards efficient solutions of bitruss decomposition for large-scale bipartite graphs. *VLDB J.* 31, 2 (2022), 203–226.
- [45] Kai Wang, Xuemin Lin, Lu Qin, Wenjie Zhang, and Ying Zhang. 2023. Accelerated butterfly counting with vertex priority on bipartite graphs. *VLDB J.* 32, 2 (2023), 257–281.
- [46] Yue Wang, Ruiqi Xu, Xun Jian, Alexander Zhou, and Lei Chen. 2022. Towards Distributed Bitruss Decomposition on Bipartite Graphs. *Proc. VLDB Endow.* 15, 9 (2022), 1889–1901.
- [47] Zhibin Wang, Longbin Lai, Yixue Liu, Bing Shui, Chen Tian, and Sheng Zhong. 2023. I/O-Efficient Butterfly Counting at Scale. *Proc. ACM Manag. Data* 1, 1 (2023), 34:1–34:27.
- [48] Yao Wu, Xudong Liu, Min Xie, Martin Ester, and Qing Yang. 2016. CCCF: Improving Collaborative Filtering via Scalable User-Item Co-Clustering. In *WSDM*. 73–82.
- [49] Feng Xia, Shuo Yu, Chengfei Liu, Jianxin Li, and Ivan Lee. 2022. CHIEF: Clustering With Higher-Order Motifs in Big Networks. *IEEE Trans. Netw. Sci. Eng.* 9, 3 (2022), 990–1005.
- [50] Dongkuan Xu, Wei Cheng, Bo Zong, Jingchao Ni, Dongjin Song, Wenchao Yu, Yuncong Chen, Haifeng Chen, and Xiang Zhang. 2019. Deep Co-Clustering. In *SDM*. 414–422.
- [51] An Yan, Zhankui He, Jiacheng Li, Tianyang Zhang, and Julian John McAuley. 2023. Personalized Showcases: Generating Multi-Modal Explanations for Recommendations. In *SIGIR*. 2251–2255.
- [52] Jianye Yang, Yun Peng, Dian Ouyang, Wenjie Zhang, Xuemin Lin, and Xiang Zhao. 2023. (p, q)-biclique counting and enumeration for large sparse bipartite graphs. *VLDB J.* 32, 5 (2023), 1137–1161.
- [53] Jianye Yang, Yun Peng, and Wenjie Zhang. 2021. (p, q)-biclique Counting and Enumeration for Large Sparse Bipartite Graphs. *Proc. VLDB Endow.* 15, 2 (2021), 141–153.
- [54] Renchi Yang and Jieming Shi. 2024. Efficient High-Quality Clustering for Large Bipartite Graphs. *Proc. ACM Manag. Data* 2, 1 (2024), 23:1–23:27.
- [55] Yixing Yang, Yixiang Fang, Maria E. Orlowska, Wenjie Zhang, and Xuemin Lin. 2021. Efficient Bi-triangle Counting for Large Bipartite Networks. *Proc. VLDB Endow.* 14, 6 (2021), 984–996.
- [56] Yichun Yang, Rong-Hua Li, Meihao Liao, Longlong Lin, and Guoren Wang. 2024. Scaling Up Graph Propagation Computation on Large Graphs: A Local Chebyshev Approximation Approach. *CoRR* abs/2412.10789 (2024).
- [57] O. N. Yaveroglu, N. Malod-Dognin, D. Davis, Z. Levnajic, V. Janjic, R. Karapandza, A. Stojmirovic, and N. Przulj. 2014. Revealing the Hidden Language of Complex Networks. *Scientific Reports* 4 (2014).
- [58] Xiaowei Ye, Rong-Hua Li, Qiangqiang Dai, Hongchao Qin, and Guoren Wang. 2023. Efficient Biclique Counting in Large Bipartite Graphs. *Proc. ACM Manag. Data* 1, 1 (2023), 78:1–78:26.
- [59] Xiaowei Ye, Rong-Hua Li, Longlong Lin, Shaojie Qiao, and Guoren Wang. 2024. Fast Biclique Counting on Bipartite Graphs: A Node Pivot-based Approach. *arXiv preprint arXiv:2412.16485* (2024).
- [60] Hao Yin, Austin R. Benson, Jure Leskovec, and David F. Gleich. 2017. Local Higher-Order Graph Clustering. In *KDD*. 555–564.
- [61] Yi Zheng, Hongchao Qin, Jun Zheng, Fusheng Jin, and Rong-Hua Li. 2020. Butterfly-Based Higher-Order Clustering on Bipartite Networks. In *KSEM*,

- Vol. 12274. 485–497.
- [62] Dawei Zhou, Si Zhang, Mehmet Yigit Yildirim, Scott Alcorn, Hanghang Tong, Hasan Davulcu, and Jingrui He. 2017. A Local Algorithm for Structure-Preserving Graph Cut. In *KDD*. 655–664.
  - [63] Dawei Zhou, Si Zhang, Mehmet Yigit Yildirim, Scott Alcorn, Hanghang Tong, Hasan Davulcu, and Jingrui He. 2021. High-Order Structure Exploration on Massive Graphs: A Local Graph Clustering Perspective. *ACM Trans. Knowl. Discov. Data* 15, 2 (2021), 18:1–18:26.
  - [64] Junjie Zhou, Meihao Liao, Rong-Hua Li, Longlong Lin, and Guoren Wang. 2025. One Index for All: Towards Efficient Personalized PageRank Computation for Every Damping Factor. *Proc. ACM Manag. Data* 3, 4 (2025), 258:1–258:26.
  - [65] Zhixin Zhou and Arash A. Amini. 2019. Analysis of spectral clustering algorithms for community detection: the general bipartite setting. *J. Mach. Learn. Res.* 20 (2019), 47:1–47:47.
  - [66] Zhixin Zhou and Arash A. Amini. 2020. Optimal Bipartite Network Clustering. *J. Mach. Learn. Res.* 21 (2020), 40:1–40:68.
  - [67] Katharina Anna Zweig and Michael Kaufmann. 2011. A systematic approach to the one-mode projection of bipartite graphs. *Soc. Netw. Anal. Min.* 1, 3 (2011), 187–218.

## A Appendix

### A.1 Further Related Work

**Bipartite Graph Clustering.** Clustering in bipartite graphs is notably more complex, yet offers greater utility due to their unique structure and diverse applications [2, 5, 65, 66]. A common method for clustering bipartite graphs involves projecting a bipartite graph  $G = (L, R, E)$  onto a unipartite graph  $\bar{G} = (L, E_L)$  (or  $\bar{G} = (R, E_R)$ ), where vertices in  $L$  (or  $R$ ) are connected if they share common neighbors in  $G$ . Once this projection is completed, traditional graph clustering algorithms can be applied to  $\bar{G}$  [39, 67]. Recent advancements focus on the simultaneous clustering, also known as co-clustering, of both vertex sets  $L$  and  $R$  in bipartite graphs [1, 7, 8, 22]. For instance, [7, 22] extended spectral clustering techniques to accommodate bipartite structures. Other efforts have sought to extend the concept of modularity [32] in unipartite graphs to bipartite graphs, leading to the development of bi-modularity for clustering both vertex sets simultaneously [1, 5, 23]. However, these modularity-based approaches suffer from a resolution limit [14], which prevents them from accurately detecting small clusters with high modularity scores. Summing up, all these methods mainly focus on the internal or external lower-order edges within clusters, making them unable to capture higher-order structural information (e.g., bicliques) necessary for more nuanced graph clustering.

**Higher-order Graph Clustering.** Motifs (i.e., small yet significant subgraphs) play a crucial role in uncovering the higher-order structure of complex networks [31]. Consequently, numerous motif-based higher-order graph clustering methods have been developed in the literature. Notable examples include motif conductance [6, 41, 60, 62, 63], motif modularity [4, 20], and the higher-order densest subgraph model [12, 38]. For instance, Benson et al. [6] introduced the concept of motif conductance in their seminal *Science* 2016 paper, demonstrating how it enhances graph clustering by accounting for the number of motifs (rather than just edges) being cut. Motif modularity extends traditional modularity by optimizing the difference between the fraction of motif instances within a cluster and the expected fraction in a random network with the same degree distribution [4, 20]. The higher-order densest subgraph model defines density as the number of motif instances divided by the number of vertices in the subgraph [12, 38]. However, as these methods are not specifically designed for bipartite graphs, they fail

to capture the unique properties of bipartite structures, leading to suboptimal performance, as stated in our experiments (Section 4).

### A.2 Basic Solutions

**A.2.1 BFV [6].** Let  $\mathcal{A}$  be the biclique adjacency matrix, where  $\mathcal{A}_{uv} = W_{uv}$  for all  $u, v \in L \cup R$  (see Section 3.1.1 for  $W_{uv}$ ). Let  $\mathcal{D}$  be the diagonal degree matrix, defined by  $\mathcal{D}_{uu} = \sum_v \mathcal{A}_{uv}$  for all  $u \in L \cup R$ . The normalized Laplacian matrix corresponding to the biclique adjacency matrix is then given by  $\mathcal{L} = I - \mathcal{D}^{-1/2} \mathcal{A} \mathcal{D}^{-1/2}$ , where  $I$  is the identity matrix. Based on these symbols, the *Science* paper [6] introduced the well-known higher-order spectral clustering method, which is detailed in Algorithm 2 (note that [6] addresses unipartite graphs and arbitrary motifs, and therefore can directly handle our problem of biclique-preserving clustering). The process begins by computing the normalized Laplacian matrix  $\mathcal{L}$  by enumerating  $(p, q)$ -bicliques (Lines 1-6). Subsequently, the algorithm identifies the eigenvector  $x$  corresponding to the second smallest eigenvalue of  $\mathcal{L}$  to perform the sweep procedure (Lines 7-9). This involves sorting all entries of  $x$  in descending order such that  $x_1 \geq x_2 \geq \dots \geq x_n$ , and then selecting the set  $S = \arg \min \phi_{\mathcal{G}^{\mathbb{B}pq}}(S_i)$ , where  $S_i = \{x_1, x_2, \dots, x_i\}$  and  $\phi_{\mathcal{G}^{\mathbb{B}pq}}(S_i)$  represents the traditional edge-based conductance [28, 40] of  $S_i$  on the weighted graph  $\mathcal{G}^{\mathbb{B}pq}$ . Namely,  $\phi_{\mathcal{G}^{\mathbb{B}pq}}(S_i) = \frac{\sum_{u \in S_i, v \notin S_i} W_{uv}}{\min\{\sum_{u \in S_i} \mathcal{D}_{uu}, \sum_{v \notin S_i} \mathcal{D}_{vv}\}}$ . In Lines 10-13, the algorithm outputs the smaller set between  $S$  and its complement  $\bar{S}$ , since  $\phi_{\mathcal{G}^{\mathbb{B}pq}}(S) = \phi_{\mathcal{G}^{\mathbb{B}pq}}(\bar{S})$ . The following theorems provide key theoretical foundations for Algorithm 2.

**THEOREM 4 ([6]).** *Given an unweighted and undirected bipartite graph  $G$ , and two positive integer parameters  $p$  and  $q$ , for any cluster  $G(X, Y)$ , we have*

$$\phi_{pq}(X, Y) = \begin{cases} \phi_{\mathcal{G}^{\mathbb{B}pq}}(X \cup Y), & \text{if } p + q = 3 \\ \phi_{\mathcal{G}^{\mathbb{B}pq}}(X \cup Y) - \frac{\sum_{B \in \mathbb{B}pq} I(|B \cap (X \cup Y)|=2)}{\sum_{u \in X \cup Y} \mathcal{D}_{uu}}, & \text{if } p + q = 4 \\ \neq \phi_{\mathcal{G}^{\mathbb{B}pq}}(X \cup Y), & \text{if } p + q > 4 \end{cases} \quad (8)$$

$I(\cdot)$  is the indicator function and  $\phi_{pq}(X, Y) \neq \phi_{\mathcal{G}^{\mathbb{B}pq}}(X \cup Y)$  indicates the relationship between  $\phi_{pq}(X, Y)$  and  $\phi_{\mathcal{G}^{\mathbb{B}pq}}(X \cup Y)$  is unclear.

**THEOREM 5 (CHEEGER INEQUALITY [3]).** *Given a weighted graph  $\mathcal{G}^{\mathbb{B}pq}$  and its normalized Laplacian matrix  $\mathcal{L}$ , we assume that  $\lambda_2$  is the second smallest eigenvalue of  $\mathcal{L}$ . Then, we have  $\lambda_2/2 \leq \phi^* \leq \sqrt{2\lambda_2}$ , in which  $\phi^*$  is the optimal edge-based conductance in terms of  $\mathcal{G}^{\mathbb{B}pq}$ .*

**THEOREM 6.** *Given an unweighted and undirected bipartite graph  $G$ , and two positive integer parameters  $p$  and  $q$  with  $p + q = 3$ . Let  $S$  be the vertex subset returned by Algorithm 2, we have  $\phi_{pq}^* \leq \phi_{pq}(S) \leq 2\sqrt{\phi_{pq}^*}$ , in which  $\phi_{pq}^*$  is the optimal biclique conductance.*

**PROOF.** Since  $S$  is equivalent to the vertices return by Fielder vector-based spectral clustering over the motif-based weighted graph  $\mathcal{G}^{\mathbb{B}pq}$  (Algorithm 2), we can obtain  $\phi_{\mathcal{G}^{\mathbb{B}pq}}(S) \leq \sqrt{2\lambda_2}$  [13, 28], where  $\lambda_2$  is the second smallest eigenvalue of the normalized Laplacian matrix of  $\mathcal{G}^{\mathbb{B}pq}$ . Meanwhile, by Theorem 5, we have  $\phi_{\mathcal{G}^{\mathbb{B}pq}}(S) \leq \sqrt{2\lambda_2} \leq \sqrt{2 \times 2 \times \phi^*} = 2\sqrt{\phi^*}$ , in which  $\phi^*$  is the optimal edge-based conductance in terms of the weighted graph  $\phi_{pq}$ . Furthermore, by Theorem 4, we have  $\phi_{pq}(H) = \phi_{\mathcal{G}^{\mathbb{B}pq}}(H)$  for any

**Algorithm 4** BPR [26]

---

**Input:** A bipartite  $G = (L, R, E)$  and two parameters  $p$  and  $q$   
**Output:** A biclique-preserving cluster  $S$

- 1:  $\mathbb{B}_{pq} \leftarrow$  all the  $(p, q)$ -bicliques of  $G$
- 2: Initializing the biclique degree  $\mathbb{B}(u) = 0$  for any  $u \in L \cup R$
- 3: **for** each  $(p, q)$ -biclique  $B \in \mathbb{B}_{pq}$  **do**
- 4:     **for** each node  $u \in B$  **do**
- 5:          $\mathbb{B}(u) + = 1$
- 6:  $i \leftarrow 1$ ;  $S_i \leftarrow L \cup R$ ;  $\mathbb{B}_{p+q}^{S_i}(u) = \mathbb{B}(u)$  and  $\mathbb{B}_1^{S_i}(u) = 0$  for  $u \in S_i$
- 7:  $Br_{S_i}(u) \leftarrow \frac{\mathbb{B}(u) + \mathbb{B}_{p+q}^{S_i}(u) - \mathbb{B}_1^{S_i}(u)}{\mathbb{B}(u)}$  for any  $u \in S_i$
- 8: **while**  $S_i \neq \emptyset$  **do**
- 9:      $u \leftarrow \arg \min \{Br_{S_i}(u) | u \in S_i\}$
- 10:      $i \leftarrow i + 1$
- 11:      $S_i \leftarrow S_{i-1} \setminus \{u\}$
- 12:  $S \leftarrow \arg \min_{S \in \{S_1, S_2, \dots, S_n\}} \{\phi_{pq}(S) | vol_{pq}(S) \leq vol_{pq}((L \cup R) \setminus S)\}$
- 13: **return**  $S$

---

**Algorithm 2** BFV [6]

---

**Input:** A bipartite  $G$  and two parameters  $p$  and  $q$   
**Output:** A biclique-preserving cluster  $S$

- 1:  $\mathbb{B}_{pq} \leftarrow$  all the  $(p, q)$ -bicliques of  $G$
- 2: Initializing  $\mathcal{A} \in R^{n \times n}$  and  $\mathcal{D} \in R^{n \times n}$  are zero matrices
- 3: **for** each  $(p, q)$ -biclique  $B \in \mathbb{B}_{pq}$  **do**
- 4:     **for** each node  $u \in B$  and node  $v (\neq u) \in B$  **do**
- 5:          $\mathcal{A}_{uv} + = 1$ ;  $\mathcal{A}_{vu} + = 1$ ;  $\mathcal{D}_{uu} + = 1$ ;  $\mathcal{D}_{vv} + = 1$
- 6:  $\mathcal{L} \leftarrow I - \mathcal{D}^{-1/2} \mathcal{A} \mathcal{D}^{-1/2}$
- 7:  $x \leftarrow$  eigenvector of the second smallest eigenvalue of  $\mathcal{L}$
- 8:  $x_i \leftarrow$  to be index of  $x$  with  $i$ th largest value
- 9:  $S \leftarrow \arg \min \phi_{\mathcal{G}^{pq}}(S_i)$ , where  $S_i = \{x_1, x_2, \dots, x_i\}$  and  $\phi_{\mathcal{G}^{pq}}(S_i)$  is the traditional edge-based conductance of  $S_i$  in terms of the weighted graph  $\mathcal{G}^{pq}$  constructed by  $\mathcal{A}$
- 10: **if**  $|S| > |\bar{S}|$  **then**
- 11:     **return**  $\bar{S}$
- 12: **else**
- 13:     **return**  $S$

---

**Algorithm 3** BPP [60]

---

**Input:** A bipartite  $G$ , three parameters  $p, q$ , and  $\alpha$ , and the use-initiated seed node  $s$ .  
**Output:** A biclique-preserving cluster  $S$

- 1: Executing Lines 1-5 of Algorithm 2
- 2:  $\pi \leftarrow$  the probability distribution by Equation 9
- 3:  $y \leftarrow \pi \mathcal{D}^{-1}$
- 4:  $y_i \leftarrow$  to be index of  $y$  with  $i$ th largest non-zero value
- 5:  $S \leftarrow \arg \min \phi_{pq}(S_i)$ , where  $S_i = \{y_1, y_2, \dots, y_i\}$
- 6: **if**  $|S| > |\bar{S}|$  **then**
- 7:     **return**  $\bar{S}$
- 8: **else**
- 9:     **return**  $S$

---

vertex subset  $H \subseteq V$  due to  $p + q = 3$ . Thus,  $\phi_{pq}(S) \leq 2\sqrt{\phi_{pq}^*}$ . Clearly,  $\phi_{pq}^* \leq \phi_{pq}(S)$  due to  $\phi_{pq}^*$  is the smallest biclique conductance. As a result, this theorem is proved.  $\square$

**THEOREM 7.** Given a bipartite graph  $G$  and two parameters  $p$  and  $q$ , the time complexity and space complexity of Algorithm 2 are  $O(\text{Eum}(B_{pq}))$  and  $O(\min\{n^2, \binom{p+q}{2} |\mathbb{B}_{pq}|\} + m)$ .  $O(\text{Eum}(B_{pq}))$  is the time complexity of enumeration  $(p, q)$ -biclique instances.

**PROOF.** Algorithm 2 first takes  $O(\text{Eum}(B_{pq}))$  time to enumerate the  $(p, q)$ -bicliques (Line 1). Next, in Lines 3-5, the algorithm utilizes  $O(\binom{p+q}{2} |\mathbb{B}_{pq}|)$  time to construct the biclique-based weighted graph and the corresponding diagonal degree matrix. Following this, in Line 6, the algorithm requires  $O(\min\{n^2, \binom{p+q}{2} |\mathbb{B}_{pq}|\})$  time to compute the matrix  $\mathcal{L}$ , as the size of the biclique-based weighted graph can reach  $O(\min\{n^2, \binom{p+q}{2} |\mathbb{B}_{pq}|\})$  in the worst case. In Line 7, Algorithm 2 takes  $O(\min\{n^2, \binom{p+q}{2} |\mathbb{B}_{pq}|\})$  time to find the eigenvector corresponding to the second smallest eigenvalue of  $\mathcal{L}$  using the well-known Lanczos iteration algorithm [17]. Finally, in Line 8 and Line 9, the algorithm requires  $O(n \log n)$  time and  $O(\min\{n^2, \binom{p+q}{2} |\mathbb{B}_{pq}|\})$  time, respectively, to sort the index and to calculate the conductance of  $S_i$ . Since  $O(\min\{n^2, \binom{p+q}{2} |\mathbb{B}_{pq}|\}) < O(\text{Eum}(B_{pq}))$ , the total time complexity of Algorithm 2 is  $O(\text{Eum}(B_{pq}))$ .

For space complexity, Algorithm 2 takes  $O(\min\{n^2, \binom{p+q}{2} |\mathbb{B}_{pq}|\})$  to store matrices  $\mathcal{A}$ ,  $\mathcal{D}$ , and  $\mathcal{L}$ . Additionally, it needs  $O(m)$  space to store the input graph  $G$ . Consequently, the total space requirement for Algorithm 2 is  $O(\min\{n^2, \binom{p+q}{2} |\mathbb{B}_{pq}|\} + m)$ .  $\square$

**A.2.2 BPP [60].** BPP is local clustering methods that perform local graph diffusion starting from a given seed vertex  $s$  to identify higher-order clusters. Let  $\mathcal{P} = \mathcal{D}^{-1} \mathcal{A}$  represent the probability transition matrix of the motif-based weighted graph  $\mathcal{G}^{pq}$ . Furthermore,  $\mathcal{P}^k$  denotes the  $k$ -hop probability transition matrix of  $\mathcal{G}^{pq}$ , where  $\mathcal{P}_{uv}^k$  indicates the probability that a random walk of  $k$ -hops ( $k \geq 1$ ) starting from vertex  $u$  ends at vertex  $v$ . On the other hand, Personalized PageRank (PPR) [56, 64] is a variant of the random walk with restart. This process is guided by a teleportation parameter  $\alpha$  and operates as follows: (1) The walk begins at a specified seed vertex  $s$ . (2) At each step, the walk halts at the current vertex with probability  $\alpha$ . Otherwise, it continues to the next vertex based on the probability transition matrix  $\mathcal{P}$ . The probability that the  $\alpha$ -discounted random walk stops at vertex  $u$  is denoted as  $\pi(u)$ :

$$\pi(u) = \sum_{k=0} \alpha(1-\alpha)^k \mathcal{P}_{su}^k, \quad (9)$$

Based on these backgrounds, we introduce the Biclique-Preserving Clustering Based on Personalized PageRank (BPP) [60]. Specifically, Algorithm 3 proceeds as follows: (1) Compute the probability distribution  $\pi$  from the Personalized PageRank random walk (i.e., Equation 9). Then, calculate  $y = \pi \mathcal{D}^{-1}$  (Lines 1-3). (2) Perform a sweep procedure to identify the cluster (Lines 4-5). This involves sorting all non-zero entries of  $y$  in descending order,  $y_1 \geq y_2 \geq \dots \geq y_{\text{sup}(y)}$ , where  $\text{sup}(y)$  represents the number of non-zero entries in  $y$ . The set  $S$  is then defined as  $S = \arg \min \phi_{pq}(S_i)$ , where  $S_i = \{y_1, y_2, \dots, y_i\}$ . (3) Finally, in Lines 6-9, the algorithm outputs the smaller of  $S$  and its complement  $\bar{S}$ , taking advantage of the fact that  $\phi_{pq}(S) = \phi_{pq}(\bar{S})$ .

**THEOREM 8.** ([60]) The time and space complexity of Algorithm 3 are  $O(\text{Eum}(B_{pq}))$  and  $O(\min\{n^2, \binom{p+q}{2} |\mathbb{B}_{pq}|\} + m)$ .

**A.2.3 BPR [26].** Algorithm 4 provides the pseudo-code of the Biclique-preserving Peeling Removal (*BPR*). Specifically, Algorithm 4 first calculates the biclique resident value for each vertex (Lines 1-7). Then, Algorithm 4 involves iteratively removing the vertex with the smallest biclique resident value, a process referred to as *peeling* (Lines 8-11). Finally, Lines 12-13 outputs the result with the smallest biclique conductance observed during the peeling process.

**THEOREM 9.** ([26]) *The time complexity and space complexity of Algorithm 4 are  $O(Eum(B_{pq}))$  and  $O(m)$ .*

**THEOREM 10.** ([26]) *Algorithm 4 can identify a cluster with biclique conductance  $\frac{1}{2} + \frac{\phi_{pq}^*}{2}$ , where  $\phi_{pq}^*$  is the optimal value.*